

University Course Registration System

Project Overview

The University Course Registration System is designed to streamline the process of course enrollment for students and administrative management of course allocations. The system enables students to register for courses, manage their schedules, and handle waiting lists dynamically. Administrators can oversee course allocations, seat availability, and user management. This project was developed by a team of three using the Code::Blocks IDE.

Objectives

Streamline Course Registration:

- Simplify the process for students to register for courses, select preferred sections, and manage their course schedules.
- Ensure that the system is user-friendly, intuitive, and accessible.

Efficient Seat Management:

- Provide real-time updates on seat availability to avoid over-enrollment.
- Implement a fair and transparent waiting list system for courses that reach capacity.

Administrative Oversight:

- Enable administrators to manage course offerings, allocate seats, and oversee user accounts effectively.
- Generate reports and analytics to assist in academic planning and resource allocation.

Key Components

User Registration and Authentication:

- **Secure User Accounts:** Students and administrators create secure accounts with password protection and data encryption.
- **Role-Based Access:** Differentiate between student and administrator roles, ensuring appropriate access to features and data.

Course Selection and Enrollment:

- **Course Listings:** Provide a comprehensive list of available courses, including detailed information such as course descriptions, prerequisites, instructors, and available sections.
- **Real-Time Enrollment:** Allow students to enroll in courses, check seat availability in real-time, and receive immediate feedback on their registration status.

Add and Drop Courses:

- **Flexibility for Students:** Enable students to add or drop courses throughout the semester, adapting to their academic needs and interests.
- **Prerequisite Verification:** Ensure that students meet the necessary prerequisites before enrolling in a course.

Waiting List Management:

- **Dynamic Ranking System:** Rank students on waiting lists based on predefined criteria, such as academic performance, hackathon wins, reservation status, and completed internships.
- **Automatic Seat Allocation:** Automatically allocate seats to students on the waiting list as they become available, ensuring a fair and efficient process.

Administrative Controls:

- **Course Management:** Allow administrators to add, modify, or delete courses, update seat availability, and manage waiting lists.
- **User Management:** Enable administrators to manage student and admin accounts, including adding or removing users and resetting passwords.
- **Reporting and Analytics:** Generate detailed reports on course enrollment statistics, seat utilization, and waiting list statuses to assist in decision-making and academic planning.

Installation and Setup

Prerequisites

- **Code::Blocks IDE:** Download and install from Code::Blocks Official Website.
- **C Compiler:** Ensure you have a C compiler installed (included with Code::Blocks).
- And the required CSVs

1. core_courses.csv

- This file contains information about the core courses offered by the university, including course names, descriptions, and prerequisites.

2. course_seats.csv

- This file tracks the number of available seats in each course section, which is critical for managing course allocations and waitlists.

3. Design Thinking_waiting_list.csv

- This file contains the waiting list for the Design Thinking course, including student information and their respective points or ranking.

4. Elective_courses.csv

- This file includes information about the elective courses available to students, similar to core_courses.csv but specific to electives.

5. Applied Mathematics_waiting_list.csv

- Similar to the Design Thinking file, this contains the waiting list for the Applied Mathematics course.

6. Data Architecture_waiting_list.csv

- Contains the waiting list for the Data Architecture course.

7. Data Science_waiting_list.csv

- Contains the waiting list for the Data Science course.

8. Film Appreciation_waiting_list.csv

- Contains the waiting list for the Film Appreciation course.

9. Machine Learning_waiting_list.csv

- Contains the waiting list for the Machine Learning course.

10. Psychology_waiting_list.csv

- Contains the waiting list for the Psychology course.

11. Robotics_waiting_list.csv

- Contains the waiting list for the Robotics course.

12. Theoretical Physics_waiting_list.csv

- Contains the waiting list for the Theoretical Physics course.

13. reallocated_students.csv

- This file contains information about students who have been reallocated to different courses or sections due to changes in seat availability or other reasons after the admin initiates reallocation.

14. users.csv

- This file contains the list of students, including their details such as username, password, CGPA, and semester.

15. user_coursechoices.csv

- This file records the course choices made by students during the registration process including all the section preferences..

Usage in the Project

- **Core and Elective Course Management:** The `core_courses.csv` and `elective_courses.csv` files are used to manage and display the available courses to students.
- **Seat Allocation:** The `course_seats.csv` file is critical for tracking the number of available seats in each course section and managing over-enrollment.
- **Waiting Lists:** Each course's waiting list is managed using dedicated CSV files (e.g., `Design Thinking_waiting_list.csv`), where students are ranked based on predefined criteria.
- **Student Information:** The `users.csv` files contain essential information about the users, ensuring that the system can authenticate users and manage their course registrations effectively.
- **Course Choices:** The `user_coursechoices.csv` file records the courses that students wish to enroll in, which is then used during the seat allocation process.
- **Reallocation and Insurance:** The `reallocated_students.csv` file is used for managing students who need to be moved between courses or sections, while the `insurance.csv` file might be used for administrative tasks related to student insurance.

These files collectively support the functionality of the University Course Registration System by managing course information, student data, seat allocations, and waiting lists.

Installation Steps:

Open the Project in Code::Blocks:

- Launch Code::Blocks.

Build and Run the Project:

- Click on the Build button to compile the project.
- Click on the Run button to execute the project.

Usage:

Running the Program

- **Launch the Application:** Run the compiled program from Code::Blocks or CLion.
- **User Interface:** Follow the on-screen instructions to navigate through the system.
- **Output:** The output interface for the user will be seen in the terminal.

User Guide:

Student Registration:

- Enter required details (name, ID, password, etc.) to create a new student account.

Login:

- Use your credentials to log in as a student or administrator.

Course Selection:

- Browse available courses, select your preferences, and confirm your choices.

Add Course:

- Select additional courses and confirm their addition to your schedule.

Drop Course:

- Choose courses to drop from your current schedule.

Waiting List:

- View your position on waiting lists for full courses.

Admin Functions:

- Manage course allocations, seat availability, and system settings.

Limitations and Future Enhancements

Current Limitations

- **CGPA Verification:** The system does not verify the CGPA provided by students, which could lead to inaccuracies.
- **Course Completion Checks:** The system does not verify if students have completed prerequisite courses.
- **Single Semester Operation:** The system handles course registrations for only one semester at a time.

Future Enhancements

- **Integration with Academic Records:** Enhance the system to verify CGPA and course completions through integration with academic databases.
- **Multi-Semester Support:** Extend the system to handle multi-semester planning and dependencies.
- **Enhanced User Interface:** Improve the user interface for a more intuitive experience.

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References

- **Tamil Nadu Engineering Admissions (TNEA) Official Website:** Real-world example for admission processes.
- **Geeks for Geeks:** Technical reference for programming concepts and implementation strategies.

TutorialsPoint

- [TutorialsPoint - C

Programming](<https://www.tutorialspoint.com/cprogramming/index.htm>)

- Provides detailed tutorials, reference guides, and examples covering basic to advanced C programming topics.

Cprogramming.com

- [Cprogramming.com](<https://www.cprogramming.com/>)

- A resource-rich site offering tutorials, articles, and tips for C and C++ programming.

Learn-C.org

- [Learn-C.org](<https://www.learn-c.org/>)

- An interactive tutorial platform for learning C programming through hands-on exercises and examples.

W3Schools

- [W3Schools - C Tutorial](<https://www.w3schools.in/c-tutorial/>)
- Offers a well-structured tutorial covering the basics of C programming along with examples and exercises.

C Plus Plus (cplusplus.com)

- [C Plus Plus - C Library Reference](<http://www.cplusplus.com/reference/clibrary/>)
- Provides a reference guide for the C standard library functions, useful for understanding and using library functions effectively.

Stack Overflow

- [Stack Overflow - C Programming Questions](<https://stackoverflow.com/questions/tagged/c>)
- A popular community-driven platform where you can ask questions and find answers related to C programming.